Year 10 Mathematics

Graphs 2

8.3 Speed-Time Graphs

Speed-Time Graphs

STARTER · 5 MIN

Date: _____ **Title:** Speed-Time Graphs

LEARNING OBJECTIVES

- Read information from a speed-time graph
- Find acceleration from the gradient
- Find distance travelled from the area under the graph

STARTER *Answer in your book. No calculators.*

1. Work out $20 \div 10$
2. Find the area of a triangle with base 10 and height 20
3. Find the area of a rectangle 30 by 20
4. Work out $\frac{1}{2} \times (30 + 60) \times 20$

Starter — Answers

STARTER

Tick or correct in green pen

1. $20 \div 10 = \mathbf{2}$
2. $\frac{1}{2} \times 10 \times 20 = \mathbf{100}$
3. $30 \times 20 = \mathbf{600}$
4. $\frac{1}{2} \times 90 \times 20 = \mathbf{900}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can read information from a speed-time graph	☐
SILVER	I can find acceleration from the gradient	☐
GOLD	I can find distance travelled from the area under the graph	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Speed-Time Graphs

INTRODUCTION · 3 MIN

A speed-time graph shows how the speed of an object changes.

Sloping up

Speeding up (accelerating).

Gradient = acceleration

Horizontal

Constant speed.

Acceleration = 0

Sloping down

Slowing down (decelerating).

Gradient is negative

Distance

Area under the graph.

Split into triangles and rectangles

Examples 1 and 2 — Reading a Speed-Time Graph

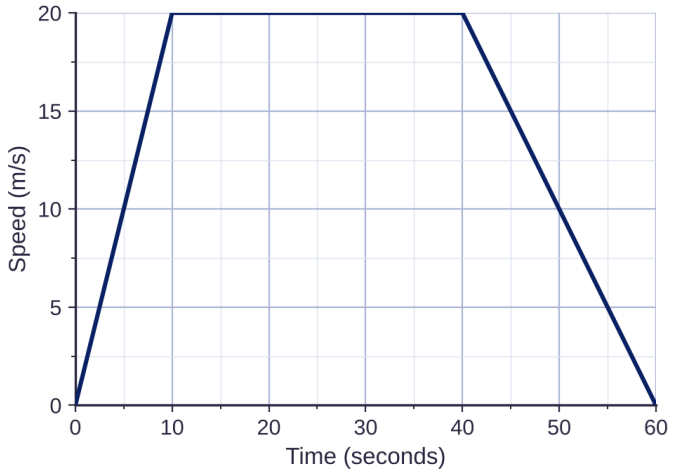
EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

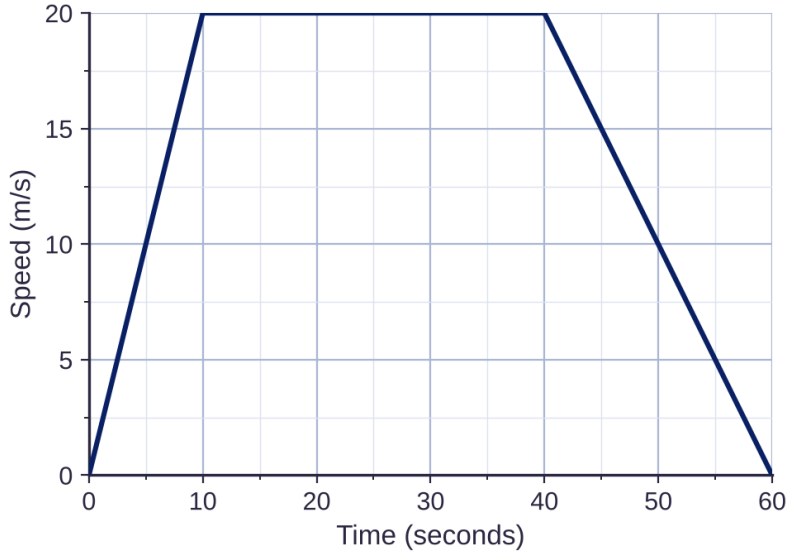
Example 1

A car starts from rest. What is its speed after 10 seconds?



Example 2

For how long does the car travel at a constant speed?



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Read up from 10 s

Speed **= 20 m/s**

Tip: read the speed axis carefully.

Example 2

Horizontal from 10 s to 40 s

40 − 10 **= 30 seconds**

Tip: constant speed is a flat line.

Task 1

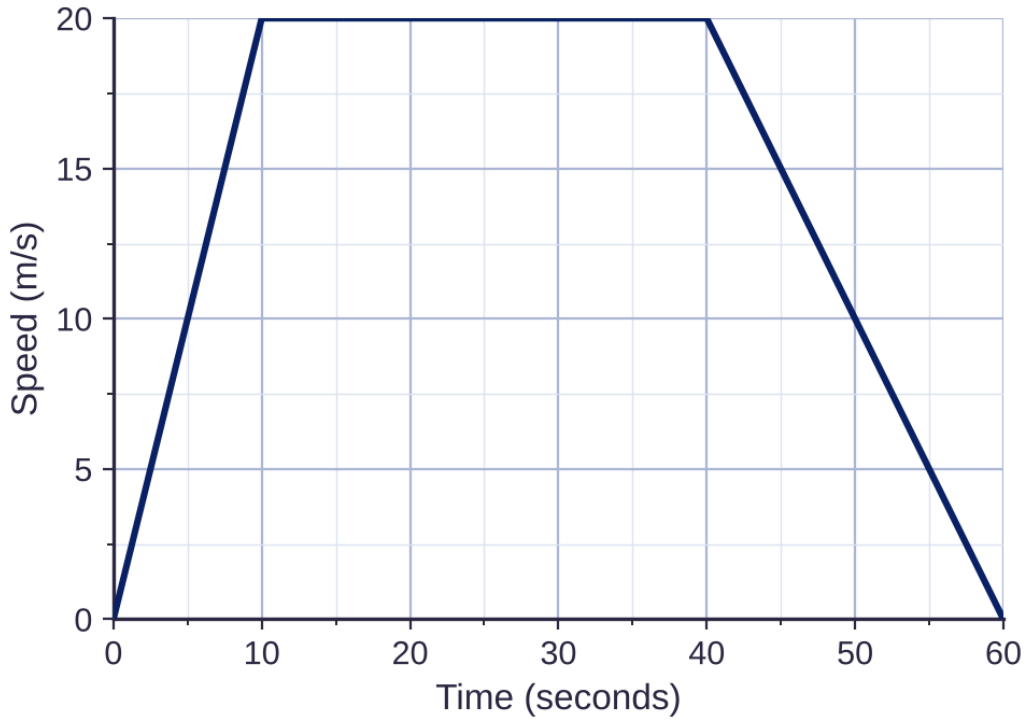
PRACTICE · 9 MIN

BRONZE

Use the speed-time graph of the car

1. What is the car's maximum speed?
2. What is the speed after 5 seconds?
3. What is the speed after 50 seconds?
4. When does the car start to slow down?
5. How long does the whole journey take?

★ Describe the motion between 40 s and 60 s.



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. Maximum speed = **20 m/s**
2. 5 seconds = **10 m/s**
3. 50 seconds = **10 m/s**
4. **Slows down at 40 seconds**
5. Total time = **60 seconds**

★ **It slows down steadily to rest**

Examples 3 and 4 — Acceleration

EXAMPLES · 5 MIN

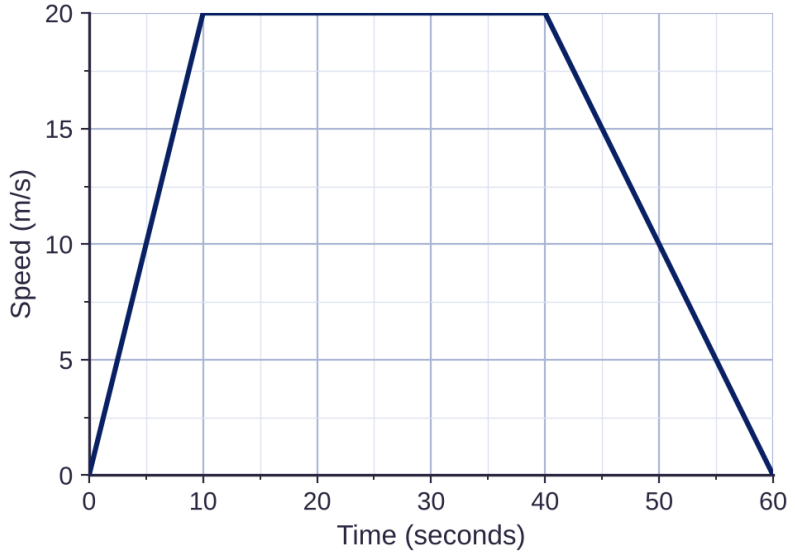
SILVER

Watch and copy into your book

Hint acceleration = $\frac{\text{change in speed}}{\text{time}}$

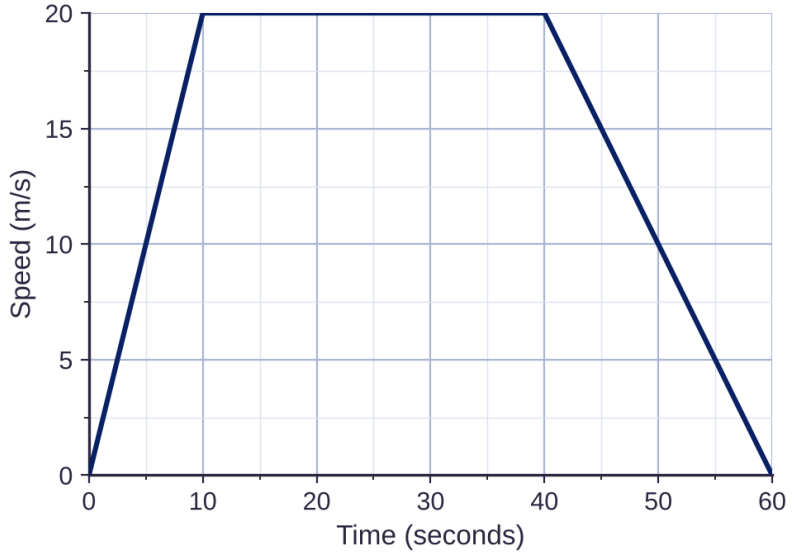
Example 3

Find the acceleration in the first 10 seconds.



Example 4

Find the deceleration between 40 s and 60 s.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$\text{Acceleration} = \frac{20}{10}$$

$$= \mathbf{2\ m/s^2}$$

Tip: acceleration = change in speed ÷ time.

Example 4

$$\text{Gradient} = \frac{-20}{20} = -1$$

Deceleration **= 1 m/s²**

Tip: a negative gradient means slowing down.

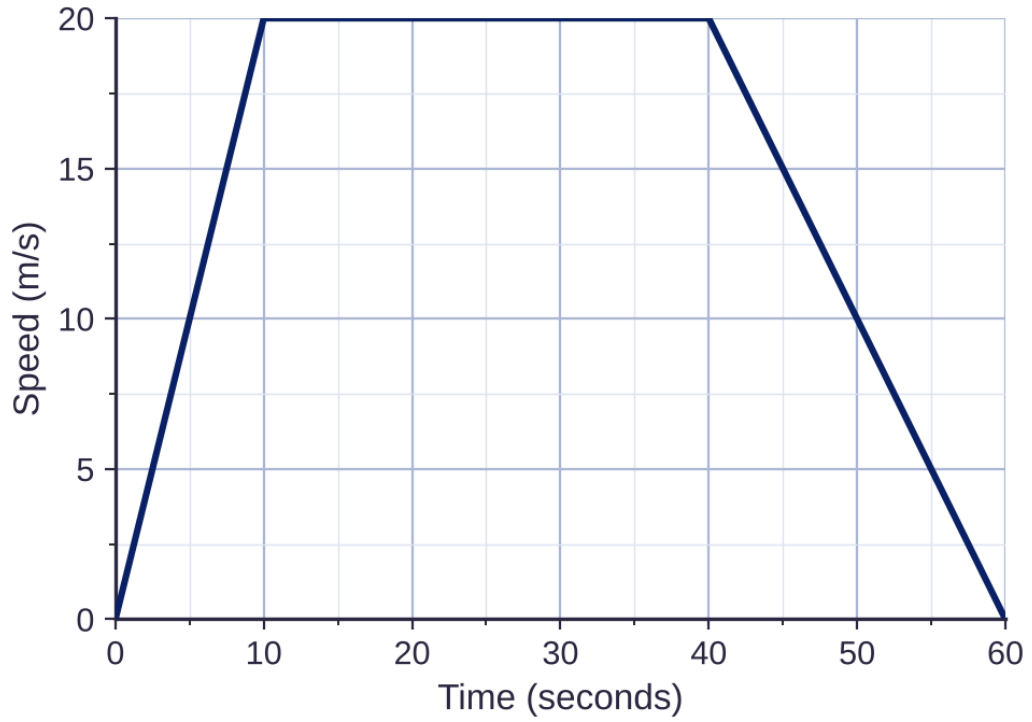
Task 2

PRACTICE · 9 MIN

SILVER *Answer in your book*

Hint acceleration = $\frac{\text{change in speed}}{\text{time}}$

1. Find the acceleration of the car between 10 s and 40 s.
2. A bike speeds up from 0 to 12 m/s in 4 s. Find its acceleration.
3. A train speeds up from 5 m/s to 25 m/s in 10 s. Find its acceleration.
4. A car slows from 30 m/s to 0 in 6 s. Find its deceleration.
5. A runner accelerates at 1.5 m/s^2 for 4 s from rest. Find her final speed.



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. Constant speed, acceleration = **0 m/s²**

2. $12 \div 4 =$ **3 m/s²**

3. $(25 - 5) \div 10 =$ **2 m/s²**

4. $30 \div 6 =$ **5 m/s²**

5. $1.5 \times 4 =$ **6 m/s**

Examples 5 and 6 — Distance Travelled

EXAMPLES · 5 MIN

GOLD

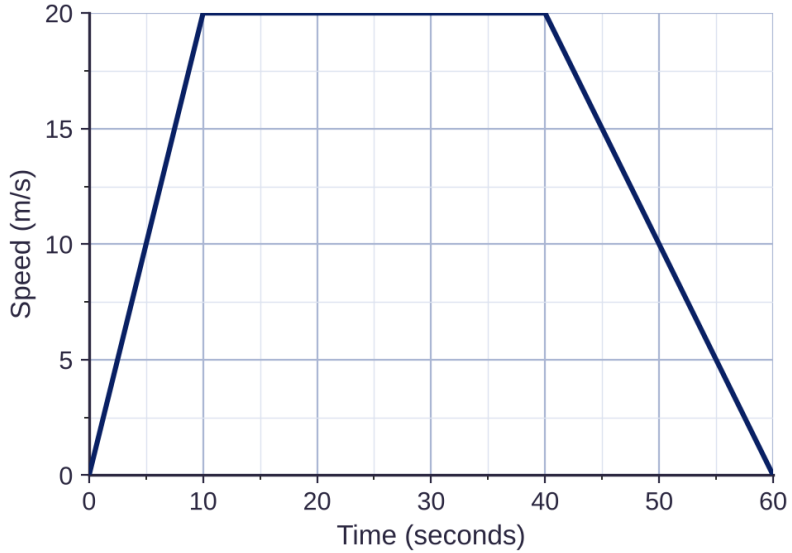
Exam-style: show all your working

Hint distance = area under the graph
acceleration = gradient

Example 5

(3 marks)

Find the total distance travelled by the car.



Example 6

(2 marks)

Find the distance travelled in the first 10 seconds.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$\text{Area} = \frac{1}{2} \times (30 + 60) \times 20$$

$$= \frac{1}{2} \times 90 \times 20$$

Distance **= 900 m**

Tip: the shape is a trapezium.

Example 6

$$\text{Area} = \frac{1}{2} \times 10 \times 20$$

Distance **= 100 m**

Tip: this part is a triangle.

Task 3

PRACTICE · 9 MIN

GOLD *Use the speed-time graph*

Hint distance = area under the graph acceleration = gradient

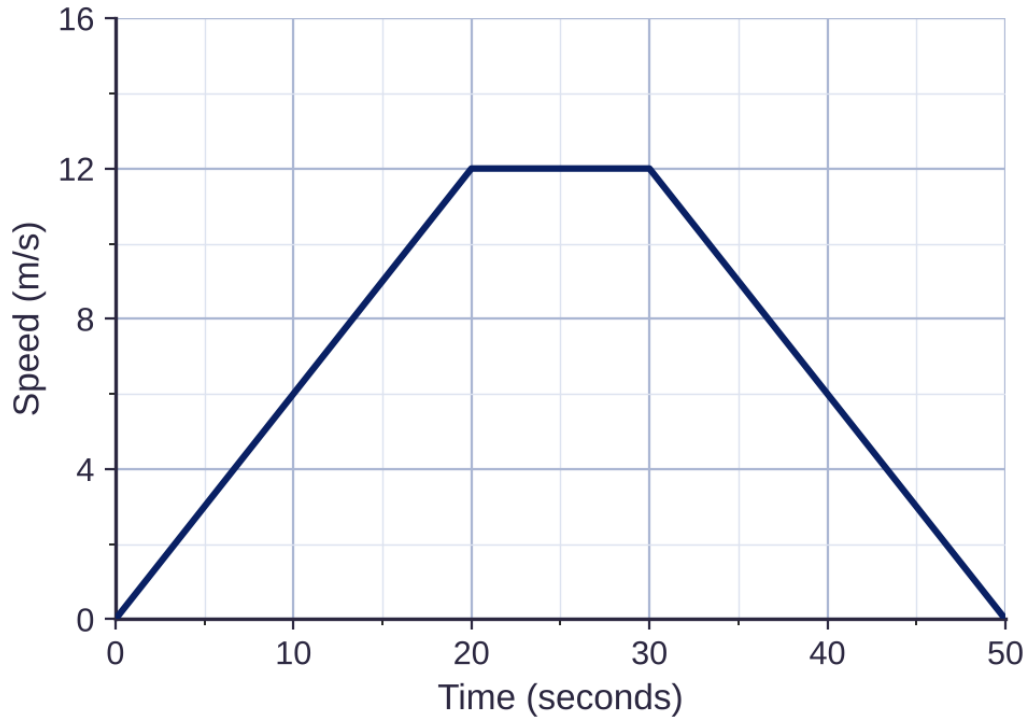
1. Find the acceleration in the first 20 seconds. (2 marks)

2. Find the distance travelled in the first 20 seconds. (2 marks)

3. Find the distance travelled between 20 s and 30 s. (2 marks)

4. Find the total distance travelled. (3 marks)

5. Find the average speed for the whole journey. (2 marks)



Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $12 \div 20 = \mathbf{0.6\ m/s^2}$

2. $\frac{1}{2} \times 20 \times 12 = \mathbf{120\ m}$

3. $10 \times 12 = \mathbf{120\ m}$

4. $\frac{1}{2} \times (10 + 50) \times 12 = \mathbf{360\ m}$

5. $360 \div 50 = \mathbf{7.2\ m/s}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE What does a horizontal line on a speed-time graph mean?

SILVER A car goes from 0 to 24 m/s in 8 s. Find the acceleration.

GOLD Find the distance: 0 to 10 m/s in 5 s, then 10 m/s for 20 s.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **Constant speed**

SILVER $24 \div 8 = \mathbf{3\ m/s^2}$

GOLD $\frac{1}{2} \times 5 \times 10 + 20 \times 10 = 25 + 200 = \mathbf{225\ m}$

Exam Practice — Reading Speed-Time Graphs

EXAM PRACTICE · 10 MIN

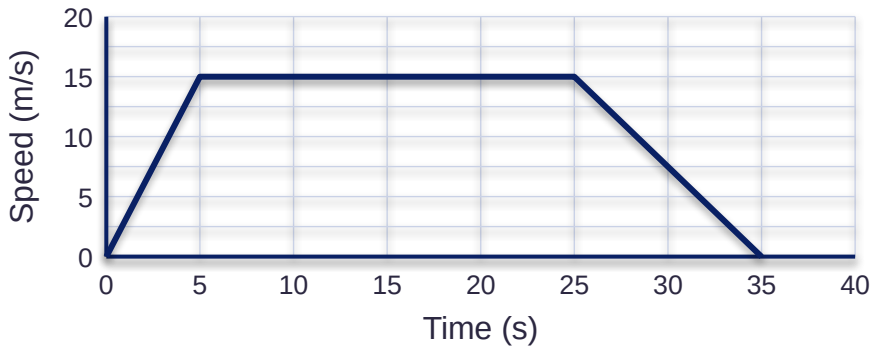
BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

The graph shows a car's journey.



(a) Write down the speed at 5 s. (1)

..... m/s

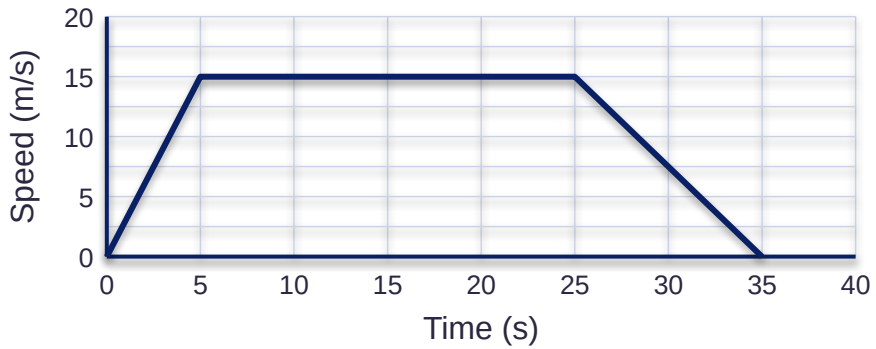
(b) Find the time at constant speed. (1)

..... s

Question 2

(2 marks)

The graph shows a car's journey.



(a) Write down the speed at 30 s. (1)

..... m/s

(b) Describe the motion after 25 s. (1)

.....

Exam Practice — Acceleration

EXAM PRACTICE · 10 MIN

SILVER

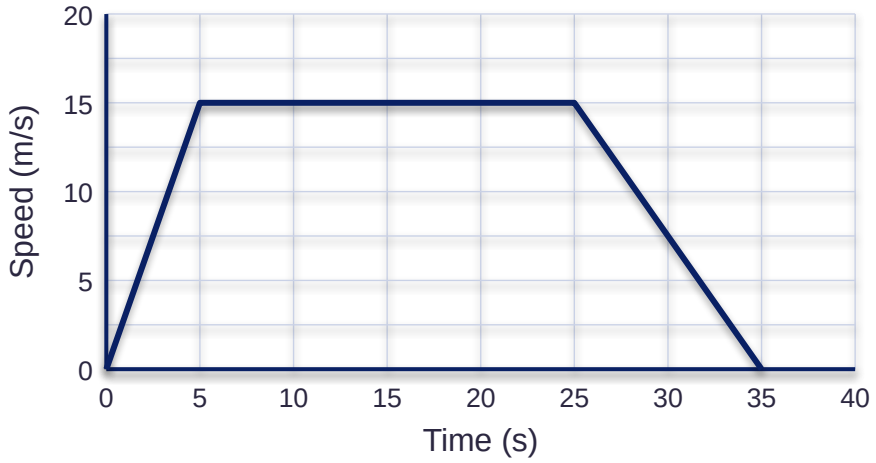
IGCSE exam-style: show all your working

Hint acceleration = $\frac{\text{change in speed}}{\text{time}}$

Question 3

(2 marks)

Work out the acceleration in the first 5 seconds.

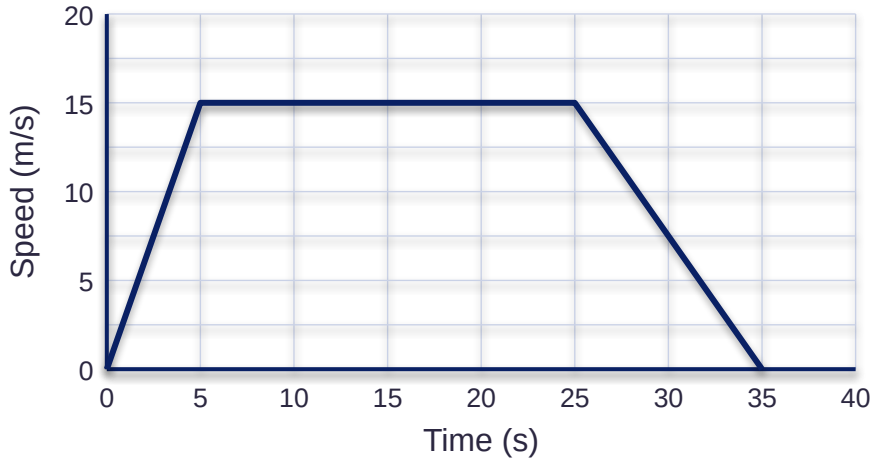


..... m/s^2

Question 4

(2 marks)

Work out the deceleration from 25 s to 35 s.



..... m/s^2

Exam Practice — Distance Travelled

EXAM PRACTICE · 10 MIN

GOLD

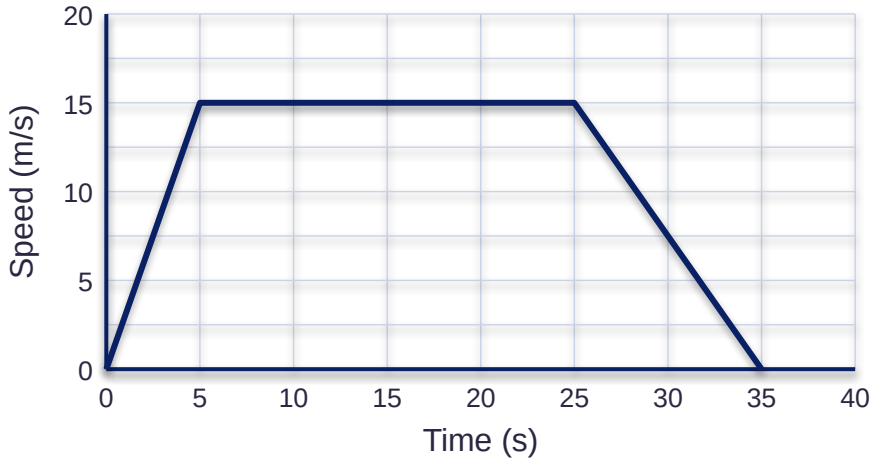
IGCSE exam-style: show all your working

Hint distance = area under the graph
acceleration = gradient

Question 5

(3 marks)

Work out the total distance travelled by the car.



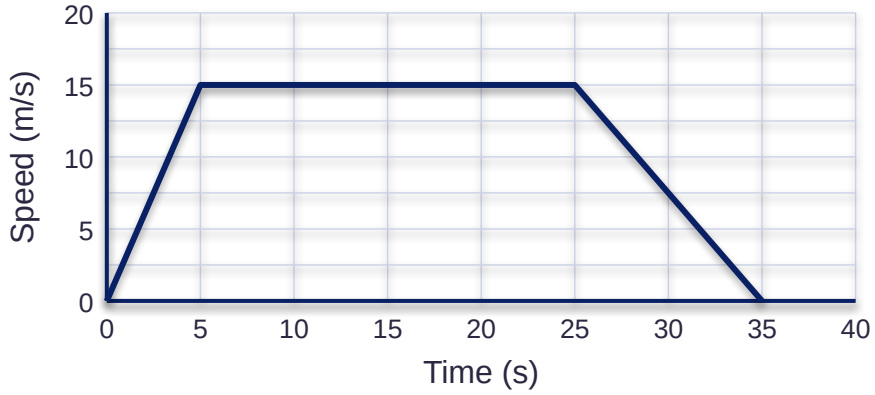
..... m

Question 6

(2 marks)

Work out the car's average speed for the whole journey.

Give your answer to 3 s.f.



..... m/s

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 15 m/s — B1

(b) 25 − 5 = 20 s — B1

Tip: flat line = constant speed, not stopped!

Question 2

(a) 7.5 m/s — B1

(b) Slows down steadily to rest — B1

Tip: compare with distance-time graphs.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

15 ÷ 5 — M1

= 3 m/s² — A1

Tip: acceleration = gradient.

Question 4

15 ÷ 10 — M1

= 1.5 m/s² — A1

Tip: deceleration is given as a positive value.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$\frac{1}{2} \times 5 \times 15 = 37.5$ — M1

$20 \times 15 = 300$

$\frac{1}{2} \times 10 \times 15 = 75$ — M1

total = 412.5 m — A1

Tip: distance = area under the graph.

Question 6

$412.5 \div 35$ — M1

= 11.8 m/s — A1

Tip: total distance ÷ total time.